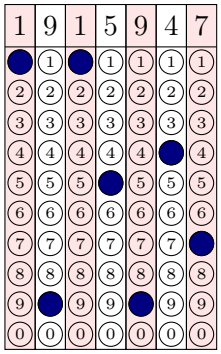
TEST VARIANTI RAQAMI: **1915947**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



1. a ning 18% i 60 ning $\frac{3}{5}$ qismiga teng. a ning $\frac{1}{5}$ qismi 15 sonidan qanchaga ko‘p?

A) *25* B) *15* C) *20* D) *10*

2. Magazinda birinchi kuni 76% tarvuz sotildi. Ikkinchi kuni esa qolgan 54 ta tarvuz sotildi. Birinchi kuni nechta tarvuz sotilgan?

A) *163* B) *171* C) *168* D) *176*

3. $x \cdot (x+7)^2 \geq 5x^2$ tengsizlikning $(1;7)$ oraliqdagi butun yechimlari yig‘indisini toping.

A) *14* B) *20* C) *28* D) *21*

4. Hisoblang:

$\cos^2 47^\circ + \cos^2 73^\circ + \cos 47^\circ \cdot \cos 73^\circ + \frac{1}{2}$.

A) $\frac{2}{3}$ B) $\frac{1}{2}$ C) 1 D) $\frac{5}{4}$

5. $7 \cdot 3^{x+1} - 5^{x+2} = 3^{x+4} - 5^{x+3}$ tenglamaning ildizi quyidagi oraliqlardan qaysi biriga tegishli?

A) $(0;2]$ B) $(-1;0]$ C) $(0;1]$ D) $(-2;-1]$

6. Ifodani soddalashtiring $(x>0)$:

$$\sqrt[4]{\frac{7-2\sqrt{6}}{3x}} \cdot \left(3\sqrt{2x} + \sqrt{3x}\right)^{0,5}$$

A) $\sqrt{3}$ B) *2* C) $\sqrt{5}$ D) *5*

7. $\left(\left(a^{-2}\right)^{-3} \cdot \left(a^3\right)^{-1} \cdot a^{-2}\right)^3 : a$ ifodaning $a = -\frac{1}{6}$ dagi qiymatini toping.

A) $\frac{1}{36}$ B) $-\frac{1}{6}$ C) 36 D) -6

8. $\left(x^3 + 35x\right)^2 = 144x^4$ tenglamaning natural yechimlari yig‘indisini toping.

A) *7* B) *5* C) *12* D) *35*

9. A, B va C sonli to‘plamlarning elementlari soni mos ravishda 10; 12 va 15 ta. $A \cup B \cup C$ to‘plamning elementlari soni eng kamida necha bo‘la oladi?

A) *12* B) *15* C) *22* D) *27*

10. $\log_3^2(x-1) - 2\log_3(x-1) > 3$ tengsizlikning barcha haqiqiy yechimlari to‘plamini toping.

A) $\left(1;\frac{4}{3}\right) \cup (28;+\infty)$

B) $\left(1;\frac{4}{3}\right)$

C) $(28;+\infty)$

D) $\left(-\infty;\frac{4}{3}\right) \cup (28;+\infty)$

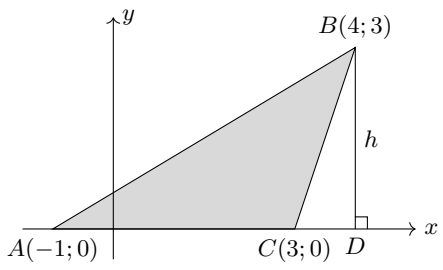
11. $3(4x-3y) - 2(-3y+4x)$ ifodani soddalashtiring.

A) $4x+3y$ B) $-4x-3y$ C) $-4x+3y$ D) $4x-3y$

12. Ketma-ket kelgan uchta tub sonlar yig‘indisi quyidagi sonlardan qaysi biriga teng bo‘lishi mumkin?

A) *15* B) *6* C) *9* D) *21*

13. Rasmda uchlari A, B va C nuqtalarda bo‘lgan uchburchakni Ox o‘qi atrofida to‘liq aylantirishdan hosil bo‘lgan jismning hajmini toping.



A) 15π B) 12π C) 6π D) 9π

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14. Tomonlari $3\sqrt{3}$; 4 va $2\sqrt{6}$ ga teng bo‘lgan uchburchakning turini aniqlang.

A) *to‘g‘ri burchakli* B) *o‘tkir burchakli*
C) *aniqlab bo‘lmaydi* D) *o‘tmas burchakli*

15. Tanga 7 marta tashlanganda 5 marta gerb va 2 marta raqam tomoni tushishining ehtimolligini toping.

A) $\frac{10}{49}$ B) $\frac{21}{128}$ C) $\frac{1}{128}$ D) $\frac{1}{32}$

16. $\left(4\frac{3}{8}-1\frac{3}{4}\right):1\frac{3}{4}$ ni hisoblang.

A) *1,25* B) *1,5* C) *0,25* D) *0,75*

17. $f(x)=13x^5+6x^3-27$ funksiya berilgan bo‘lsa, $f(f(x))$ funksiyaning darajasi toping.

A) *10* B) *15* C) *25* D) *9*

18. $y=-x+1$ va $y=x^2-5x+6$ funksiyalarning grafiklari orasidagi eng qisqa masofani toping.

A) $\frac{\sqrt{2}}{2}$ B) $\frac{\sqrt{3}}{2}$ C) $\frac{1}{2}$ D) *0*

19. $\left(x^2+\frac{1}{x^2}\right)-4\left(x+\frac{1}{x}\right)+5=0$ tenglamaning barcha haqiqiy ildizlari yig‘indisini toping.

A) *3* B) *1* C) *2* D) *4*

20. $ABCD$ parallelogrammning uchlari $A\,(-2;\,6;\,-9)$, $B\,(-12;\,6;\,5)$ va $C\,(4;\,6;\,5)$ nuqtalar bo‘lsa, $\overrightarrow{BD}$ vektorning koordinatalari yig‘indisini toping.

A) *−12* B) *−16* C) *12* D) *16*

21. $3^{\frac{1}{\log_{0,25}3}}-0,25^{1+\cos^2(5\pi-3x)}$ ifodaning eng kichik qiymatini toping.

A) 0 B) $-\frac{1}{4}$ C) $-\frac{3}{4}$ D) $\frac{3}{16}$

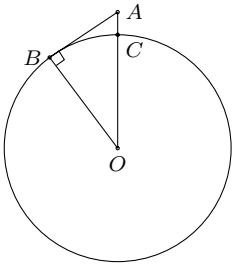
22. $f(x)=3x+b$ funksiya b ning qanday qiymat(lar)ida juft funksiya bo‘ladi?

A) $b<0$ B) $b>0$ C) $b=2n$, $n\in N$
D) *b ning hech qanday qiymatida*

23. $\int \sin x\cdot\cos^7 x\,dx$ integralni hisoblang.

A) $-\frac{1}{8}\sin^8 x+C$ B) $\frac{1}{8}\sin^8 x+C$
C) $-\frac{1}{8}\cos^8 x+C$ D) $\frac{1}{8}\cos^8 x+C$

24. Rasmda markazi O nuqtada bo‘lgan aylanaga A nuqtadan AB urinma o‘tkazilgan. AO kesma aylanani C nuqtada kesib o‘tadi. Agar aylananing kichik BC yoyi uzunligi 3 ga va radiusi 4 ga teng bo‘lsa, AC ni toping.



A) $\frac{4}{\cos 0,75}-4$ B) $4(1-\cos 0,75)$
C) $4\cos 0,75$ D) $\frac{5}{\cos 0,75}-4$

25. $f(x)=\ln\sqrt{x^2-4x+5}+3x$ funksiyaning $x_0=0$ nuqtadagi hosilasini toping.

A) $2\frac{2}{5}$ B) $2\frac{3}{5}$ C) $3\frac{2}{5}$ D) $4\frac{2}{5}$

26. Agar piramida asosining diagonallari soni bilan uning yoqlari soni yig‘indisi 22 ga teng bo‘lsa, bu piramidaning qirralar soni uchlari sonidan qanchaga ko‘p?

A) *5* B) *6* C) *8* D) *7*

27. Hisoblang: $\dfrac{(27+79)\cdot\left(2+\frac{16}{45}\right)\cdot 45^{-1}}{\left(0,(55)+\frac{1}{0,(555)}\right)^2}\cdot 0,(5)$

A) $0,(5)$ B) 9 C) 1 D) $1\frac{4}{5}$

28. x va y lar uchun $y^2+2x\,(x+y)+3\,(2x+3)=0$ tenglik o‘rinli bo‘lsa, $\frac{x^2+y^2}{6}$ ifodaning qiymatini toping.

A) *3* B) *2* C) $\frac{4}{3}$ D) *1*

29. Agar geometrik progressiyaning umumiy hadi $b_n=3\cdot 2^n$ bo‘lsa, $\frac{1}{b_1}+\frac{1}{b_2}+...+\frac{1}{b_{10}}$ yig‘indini hisoblang.

A) $\frac{341}{512}$ B) $\frac{681}{1024}$ C) $\frac{681}{512}$ D) $\frac{341}{1024}$

30. Soatning soat mili 14° burilganda minut mili necha gradus burchakka buriladi?

A) *84°* B) *360°* C) *168°* D) *720°*

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Test varianti "Test topshiriqlari to'plami 2019" asosida shakllantirilgan:

1 – 21-bet 48	7 – 23-bet 6	13 – 113-bet 53	19 – 64-bet 56	25 – 77-bet 5
2 – 19-bet 31	8 – 61-bet 21	14 – 93-bet 23	20 – 118-bet 36	26 – 107-bet 11
3 – 68-bet 18	9 – 121-bet 18	15 – 125-bet 26	21 – 40-bet 56	27 – 16-bet 63
4 – 43-bet 83	10 – 58-bet 32	16 – 11-bet 11	22 – 71-bet 10	28 – 49-bet 46
5 – 54-bet 27	11 – 45-bet 7	17 – 74-bet 42	23 – 86-bet 9	29 – 32-bet 24
6 – 27-bet 45	12 – 8-bet 11	18 – 83-bet 50	24 – 105-bet 107	30 – 97-bet 59

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